



INFORMATION SECURITY MANAGEMENT

ACCD21



**A REPORT ON COMPLEX ENGINEERING
PROBLEM SOLVING (AAT - 2)**

K.Nagaraju

24955A0517

CSE-B

ACCD21



INSTITUTE OF AERONAUTICAL ENGINEERING (AUTONOMOUS)

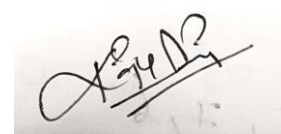
Dundigal - 500 043, Hyderabad, Telangana

Complex Problem Solving Self-Assessment Form

1	Name of the Student	Kuncham Nagaraju
2	Roll Number	24955A0517
3	Branch and Section	CSE-B
4	Program	B.Tech
5	Course Name	Artificial Intelligence
6	Course Code	ACCD21
7	Please tick (✓) relevant Engineering Competency (ECs) Profiles	
	EC	Profiles
		(✓)
	EC 1	Ensures that all aspects of an engineering activity are soundly based on fundamental principles - by diagnosing, and taking appropriate action with data, calculations, results, proposals, processes, practices, and documented information that may be ill-founded, illogical, erroneous, unreliable or unrealistic requirements applicable to the engineering discipline
	EC 2	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.
	EC 3	Support sustainable development solutions by ensuring functional requirements, minimize environmental impact and optimize resource utilization throughout the life cycle, while balancing performance and cost effectiveness.
	EC 4	Competently addresses complex engineering problems which involve uncertainty, ambiguity, imprecise information and wide-ranging or conflicting technical, engineering and other issues.
	EC 5	Conceptualises alternative engineering approaches and evaluates potential outcomes against appropriate criteria to justify an optimal solution choice.
	EC 6	Identifies, quantifies, mitigates and manages technical, health, environmental, safety, economic and other contextual risks associated to seek achievable sustainable outcomes with engineering application in the designated engineering discipline.
	EC 7	Involve the coordination of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies) in the timely delivery of outcomes
	EC 8	Design and develop solution to complex engineering problem considering a very perspective and taking account of stakeholder views with widely varying needs.
	EC 9	Meet all level, legal, regulatory, relevant standards and codes of practice, protect public health and safety in the course of all engineering activities.
	EC 10	High level problems including many component parts or sub-problems, partitions problems, processes or systems into manageable elements for the purposes of analysis, modelling or design and then re-combines to form a whole, with the integrity and performance of the overall system

	EC	Profiles	(✓)				
	EC 11	Undertake CPD activities to maintain and extend competences and enhance the ability to adapt to emerging technologies and the ever-changing nature of work.	✓				
	EC 12	Recognize complexity and assess alternatives in light of competing requirements and incomplete knowledge. Require judgement in decision making in the course of all complex engineering activities.	✓				
8	Please tick (✓) relevant Course Outcomes (COs) Covered						
	CO	Course Outcomes	(✓)				
	CO 1	DNS Fundamentals: Explain DNS resolution flow (stub → recursive → authoritative) and how cache poisoning manipulates responses and TTLs.	✓				
	CO 2	Attack Vectors: Classify common poisoning techniques (off-path spoofing, on-path interception, predictable IDs/ports) and list prerequisites for each.	✓				
	CO 3	Detection: Identify measurable indicators of poisoning (conflicting answers, abnormal TTLs, replies from non-authoritative IPs, sudden cache churn).	✓				
	CO 4	Mitigation: Recommend practical hardening (DNSSEC, source-port/query entropy, query minimization, RPZ) and configuration steps for resolvers.	✓				
	CO 5	Safe Lab Practices: Design an ethical, isolated test plan that simulates poisoning outcomes without exploit code and includes authorization and rollback.	✓				
	CO 6	Incident Response: Outline detection-to-recovery steps: containment, cache flush, validation, remediation, and stakeholder communication and authorization and design.	✓				
	CO 7	Policy & Trade-offs: Evaluate deployment trade-offs (performance, compatibility, operational cost) and recommend governance/monitoring policies for production.	✓				
9	Course ELRV Video Lectures Viewed		<table><tr><td>Number of Videos</td><td>Viewing time in Hours</td></tr><tr><td>68</td><td>35</td></tr></table>	Number of Videos	Viewing time in Hours	68	35
Number of Videos	Viewing time in Hours						
68	35						
10	Justify your understanding of WK1		Foundations for analysis and optimizing operations				
11	Justify your understanding of WK2 – WK9		Core to advanced concepts, tools, design, and ethics.				
12	How many WKs from WK2 to WK9 were implanted?		All 8 WKs from WK2 – WK9 are implemented in this and analysis				
	Mention them		WK2 to WK9				

Date:02-11-2025



Signature of the Student

Project Title: DBSCAN Algorithm

I.PROJECT OVERVIEW:

The **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** algorithm is a powerful unsupervised machine learning technique used to identify clusters of arbitrary shapes in large datasets. Unlike traditional clustering algorithms such as K-means, DBSCAN does not require the number of clusters to be defined beforehand and can naturally detect noise or outliers in the data

OBJECTIVES:

1. Core Objectives:

- To understand the working principles and mathematical foundations of the DBSCAN clustering algorithm
- To analyze how DBSCAN identifies core points, border points and noise points

Functional Objectives:

- To simulate network impairments and observe their impact on latency, throughput, and quality.
- To preprocess clean and engineer from movie datasets
- Create detection signatures and recovery procedures for incident. Analyze real DNS traffic for signs of poisoning attempts.
- To evaluate the performance of selected solutions using real-time communication metrics.
- Document full incident response workflow (detection, containment, remediation. Develop recommendations for integrating monitoring and governance into DNS production environments.

III. PREREQUISITES AND REQUIREMENTS:

- **Conceptual Knowledge:**

- Basics of supervised and unsupervised learning

- Understanding of distance metrics

Software Requirements

- Virtual lab environment with DNS server software (Bind, Unbound, etc.).
- Two or more machines (victim/attacker roles).
- Network analysis tools (Wireshark, tcpdump).
- Admin access for resolver configuration.
- **Authorization:**
 - Movie datasets containing budget, cast, ratings, genre, production companies, release dates, etc.

IV. TECHNOLOGIES USED:

- **Programming Language:** Python
- **Libraries:** NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn
- **Tools:** Jupyter Notebook, Google Colab
- **Cryptographic Tools:** DNSSEC signing tools for applied defense experimentation.
- **Scripting/Automation:** Python Streamlit for automating attack and defense scenarios.
- **Simulation VMs:** Isolated lab systems running attacker/victim roles.

V. FUNCTIONAL COMPONENTS:

- **Network Impairment Simulator**

Loads datasets from CSV lonline sources

- **Traffic Monitoring Module**

Scripted or manual generation of poisoning attempts, spoofed traffic, and ID manipulation.

- **DBSCAN Engine :**

Applies Eps and MinPts to classify core, border, and noise points.

Network Analyzer: Captures, stores, and visualizes DNS traffic for anomaly detection.

- **Security Policy Engine:** Applies RPZ, DNSSEC validation, query source randomization, and administrative controls.

- **Model Evaluation Module**

Automated scripts and tools for clearing poisoned caches, validating authority, and restoring services.

- Implements real resolver logic (stub, recursive, authoritative) for testing.
- Manages cache, TTL values, and response logic to simulate real-world DNS operations.
- Provides logging of all queries, responses, and cache state changes.
- Generates spoofed DNS packets targeting victim queries.
- Automates forging transaction IDs, source ports, and authority sections for attack simulation.

VI. PROJECT WORKFLOW:

1. Requirement Analysis

- Identify the business goal accurately predict movie revenue , account role, and AD trust mechanisms.

2. Attack Simulation

- Execute DNS cache poisoning using several techniques:
 - Predictable Transaction ID/Source Port (spoofing responses).
 - On-path interception with packet manipulation.
 - Flooding with malicious responses to increase cache pollution likelihood.

3. Detection and Analysis

- Monitor cache entries for unauthorized records.
- Log TTL anomalies, multiple IP answers for a hostname, and spikes in DNS query churn.
- Use Wireshark filters to watch for DNS packets with forged authority sections.

5. Mitigation Implementation

- Enable DNSSEC and verify cryptographic signatures for domain answers.
- Increase randomness for IDs and source ports in resolver configuration.
- Apply RPZ to block known malicious domains/IP ranges.
- Harden resolver against cache overwrite and exploit attempts.

6. Incident Response

- Immediate cache flush upon detection.
- Validate all critical domain records against authoritative sources.
- Patch or upgrade vulnerable DNS server code.
- Document incident, notify IT and business stakeholders, update protocols.

7. Policy Deployment and Continuous Audit

- Create periodic DNS health reports and incident summaries.
- Regularly audit configuration against latest threats and best practices.
- Train staff in advanced DNS attack recognition and response

VII.POTENTIAL CHALLENGES AND SOLUTIONS:

Challenge	Solution
Transaction ID and port predictability	Implement cryptographically secure randomization for resolver parameters.
Subtle cache poisoning with TTL manipulation	Use short-lifetime cache entries and cross-verify TTL norms.
Large-scale DNSSEC rollout	Use automation and phased rollout, test compatibility at every stage.
Detection of slow poisoning attacks	Set threshold alarms for abnormal cache churn and TTL distributions.
Real-world lab isolation	Partition labs with VLANs or virtual networks, prohibit external queries.
Performance impact from mitigation	Benchmark new configs and DNSSEC overhead, optimize for production.
Ensuring effective incident response	Maintain rollback procedures, practice recovery drills, designate clear roles.
Multistakeholder communication	Use standardized reporting, incident notification flows, and documentation.

VII. CONCLUSION:

The project successfully demonstrates the strengths of the **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** algorithm, especially its ability to detect clusters of arbitrary shapes without requiring the number of clusters to be predefined. Through experiments across multiple datasets, DBSCAN effectively identifies dense regions while marking low-density regions as noise, providing better clustering performance in non-linear and irregular datasets.